

Constant rates of change



- 1
- a Constant b Variable c Constant d Variable
- e Variable f Variable

2 Constant

Average rates of change

- 3
- a -4 b -3 c -7 d Variable

- 4
- a -1 b -1 c -1 d Linear

5 Non-linear

- 6
- a -4 b -4 c -4 d Linear

- 7 $-\frac{30}{\pi}$

- 8
- a 1
- b 2
- c 4
- d

Interval	$[3, 4]$	$[4, 5]$	$[5, 6]$
Average rate of change	8	16	32



Instantaneous rates of change

9 0

10

a 0

b 0

11 2

12

a 2

b 2

13

a 0

b 0

c 2

d 1

e -2

f 2

g -1

h 1

i -3

14

a 10

b 7

15

a -4

b -6.5

16

a 1

b 2

17

a -3

b -2

18

a $\frac{2}{3}$

b $\frac{1}{2}$

19

a 6

b 2

20

a -3

b -2

21

a $a = 7$

b $b = 5$



c $c = 2$

22

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	7
1	1.5	0.5	4.75
1	1.1	0.1	3.31
1	1.05	0.05	3.1525
1	1.01	0.01	3.0301
1	1.001	0.001	3.003
1	1.0001	0.0001	3.0003

b 3

23

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	6
1	1.5	0.5	5
1	1.1	0.1	4.2
1	1.05	0.05	4.1
1	1.01	0.01	4.02
1	1.001	0.001	4.002
1	1.0001	0.0001	4.0002

b 4

24

a



a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
0	1	1	3
0	0.5	0.5	2
0	0.1	0.1	1.487
0	0.05	0.05	1.4355
0	0.01	0.01	1.3959
0	0.001	0.001	0.138 73
0	0.0001	0.0001	1.3864

b 1.386

25

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	-3
1	1.5	0.5	-2.5
1	1.1	0.1	-2.1
1	1.05	0.05	-2.05
1	1.01	0.01	-2.01
1	1.001	0.001	-2.001
1	1.0001	0.0001	-2.0001

b -2

26

a



a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	-2.5
1	1.5	0.5	-3.3333
1	1.1	0.1	-4.5455
1	1.05	0.05	-4.7619
1	1.01	0.01	-4.9505
1	1.001	0.001	-4.995
1	1.0001	0.0001	-4.9995

b -5

Applications

27

a 11

b 26

c $\frac{15}{4}$

28

a 0

b 3536

c 272

29 \$300 per year

30 60

31

a $y = 4x + 20$ b $y = 20(1.10)^x$

c 4 turbines each year

d 9.59 turbines each year

e Option A

32



- a
- i \$4 per year
 - ii \$2 per year
 - iii \$6 per year
 - iv \$1 per year
- b Between 1996 and 2010

33

- a
- i -\$5 per year
 - ii -\$6 per year
 - iii -\$3 per year
 - iv -\$1 per year
- b Between 1985 and 1999

34

- a
- i \$4.30 per year
 - ii \$0.20 per year
 - iii \$6.10 per year
 - iv -\$1.40 per year
- b Between years 13 and 26.

35

- a
- i 19 km/h
 - ii 18 km/h
 - iii 11 km/h
 - iv 15 km/h
- b The first hour.

36 -\$10 per $1^\circ F$

37

a

Year	0	1	2	3	4	5
Tobia's staff	310	269	228	187	146	105
Gwen's staff	310	302	286	262	230	190

- b -41
- c -32
- d Linear, because it decreases by the same amount each year.
- e Non-linear, because it decreases by more each year.
- f Tobias's
- g Gwen's

38



a

Month	0	1	2	3	4	5
Mario's clients	0	27	54	81	108	135
Christa's clients	0	6	18	36	60	90

- b Linear
- c Non-linear
- d 27
- e 18
- f Mario's
- f Christa's

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a

Month	0	1	2	3	4	5
Value of Harry's investment	2000	2060	2120	2180	2240	2300
Value of Skye's investment	2000	2040	2080.8	2122.42	2164.87	2208.17

- b \$60
- c \$40.8
- d Linear
- e Non-linear
- f Harry's

40

a $-\frac{1}{2}$

- b The amount of water in the bucket decreases by $\frac{1}{2}$ litre every minute.

41

a $\frac{3}{5}$

- b The cost of a call increases by \$0.60 for each additional minute

42 Oliver, 10 km/h

43 6.5 insects per week

44

- a 0°F per hour b 6°F per hour c -1°F per hour d 1.5°F per hour



45

- a
i 9 m/s ii 3 m/s iii -3 m/s iv -9 m/s
- b Between $t = 0$ and $t = 1$, and between $t = 3$ and $t = 4$
- f The ball is falling down.

46

- a 130 b 114 c Product A

47

- a 1.20 b 0.37 c Fund A